EAST WEST INSTITUTE OF TECHNOLOGY

(An Autonomous Institution under VTU, Belagavi)
Off Magadi Road, Bengaluru – 560091
Department of CSE (IoT and Cyber Security including Blockchain Technology)
Academic Year 2026-27, 7th Semester
Machine Learning [BIC703] (Mini Project – Deliverables, Rubrics)

Course Instructor:	P Bharath Krishna
Mini Project Duration:	**17th Aug 2026 – 19th Oct 2026 (10 Weeks)**
Total Marks:	100 (down scaled to 10 marks for CIE)
Course Objectives:	
• Identify and formulate real-world problems that can be addressed using appropriate Machine Learning techniques. • Collect, explore, clean, and preprocess real-world datasets and justify the data preparation and feature engineering techniques applied. • Develop and compare multiple Machine Learning models and select an appropriate model based on suitable evaluation metrics. • Optimize and evaluate Machine Learning models using hyperparameter tuning and appropriate validation techniques. • Develop and deploy an end-to-end Machine Learning solution as a simple web application and document the complete ML pipeline.	
Prerequisites: Basic knowledge of Python programming, data analysis using NumPy and Pandas, data visualization, probability and statistics, Machine Learning algorithms, data preprocessing, model evaluation, and basic Git/GitHub usage**.**	

Minimum Technical Deliverables for Project

To ensure consistency and comprehensive learning, require every team to include:

- Data Set creation / identification
- Exploratory Data Analysis (minimum 5 visualizations)
- Data cleaning and preprocessing with justification
- Feature engineering or feature selection where applicable
- Comparison of at least **three applicable machine learning algorithms**
- Hyperparameter tuning for the best-performing model
- Appropriate evaluation metrics with interpretation
- Discussion of model strengths, limitations, and possible improvements
- Deployment as a simple web application using **Streamlit**, **Gradio**, or **Flask**
- Public GitHub repository containing source code, trained model, requirements file, and README
- A project report documenting the complete ML pipeline

Week wise plan / Deliverables

Week	Date	Activities	Expected Output
Week 1	17th Aug	Problem identification, domain understanding, literature/background study	Problem statement + objectives
Week 2	24th Aug	Dataset creation/collection, understanding variables	Dataset + data dictionary
Week 3	31st Aug	Exploratory Data Analysis	EDA with minimum 5 visualizations

Week 4	7th Sep	Data cleaning, preprocessing, feature engineering/selection	Cleaned dataset + preprocessing pipeline
Week 5	14th Sep	Initial ML models + Phase 1 presentation	**Phase 1 Evaluation**
Week 6	21st Sep	Model comparison and deeper experimentation	3+ ML models
Week 7	28th Sep	Hyperparameter tuning + model selection	Tuned final model
Week 8	05th Oct	Final evaluation + error analysis	Final results + interpretation
Week 9	12th Oct	Streamlit/Gradio/Flask deployment + GitHub	Working application + GitHub repository
Week 10	19th Oct	Report, presentation, demonstration and viva	**Phase 2 Evaluation**

Overall Evaluation Structure

Evaluation	**Timing**	**Marks**	**Marks Criteria**
Phase 1 – Problem Understanding & ML Pipeline	14th Sep	**40**	Evaluate problem definition, dataset, EDA, preprocessing and initial ML modelling
Phase 2 – Final Project Evaluation	19th Oct	**60**	Evaluate complete ML pipeline, tuning, evaluation, deployment, GitHub, report and viva
Total	10 Weeks	**100**	

Evaluation Rubrics

Component	**Marks**
Phase 1	
1. Problem Definition & Objectives	5
2. Dataset Creation / Collection	5
3. Exploratory Data Analysis	10
4. Data Cleaning & Preprocessing	8
5. Feature Engineering / Selection	4
6. Initial ML Model Development	5
7. Phase 1 Presentation & Viva	3
Total	**40**
Phase 2	
1. Model Comparison	8
2. Hyperparameter Tuning	8
3. Evaluation & Interpretation	8
4. Model Strengths, Limitations & Improvements	5
5. Deployment	10
6. GitHub Repository	5
7. Project Report	8
8. Final Presentation & Individual Viva	8
Total	**60**

Phase 1 Report Structure

Note: Phase 1 Report can be in the form of a PPT or Word document

Section	Contents
Project Title	Title of the ML project
Team Details	Team members, USNs, guide/faculty
Problem Statement	Real-world problem being addressed
Objectives	Specific objectives of the project
Problem Formulation	Classification / Regression / Clustering and target variable
Dataset Description	Dataset source, size, attributes, target variable
Data Dictionary	Description of important features
Exploratory Data Analysis	Analysis of data characteristics and patterns
EDA Visualizations	Minimum 5 meaningful visualizations with interpretation
Data Cleaning	Missing values, duplicates, outliers, inconsistent data, etc.
Data Preprocessing	Encoding, scaling, transformation, train-test split, etc.
Feature Engineering / Selection	Features created/selected and justification
Initial ML Models	Initial algorithms selected and rationale
Initial Results	Preliminary model performance
Challenges / Observations	Problems encountered and how they were addressed
Phase 2 Plan	Model comparison, tuning, deployment and remaining work
References	Dataset and other sources
Guide Evaluation Comments	Section to capture the Guide suggestions and evaluation

Final Report Structure

Note: Final Report can be in the form of a soft binding

Sl. No.	Chapter / Section	Contents / Subsections
1	Title Page	Project title, student names, USNs, semester, department, institution, academic year, guide name
2	Certificate / Declaration	Project completion certificate and student declaration, as applicable
3	Acknowledgement	Acknowledgement of faculty, institution, organizations, or other support
4	Abstract	Brief summary of the problem, dataset, methodology, key results, final model, and outcome
5	Table of Contents	List of chapters, sections, and page numbers
6	List of Figures and Tables	List of visualizations, diagrams, tables, and important results
7	Introduction	Background, motivation, problem statement, significance of the problem, objectives, scope, and proposed ML solution
8	Dataset Description	Dataset source, method of data collection, dataset size, attributes/features, target variable, data dictionary, and data quality

9	Exploratory Data Analysis (EDA)	Univariate, bivariate/multivariate analysis, distributions, correlations, outliers, patterns, trends, minimum 5 meaningful visualizations, and interpretation of findings
10	Data Cleaning and Preprocessing	Missing-value treatment, duplicate removal, outlier handling, categorical encoding, scaling/normalization, transformations, train-test split, and justification of preprocessing decisions
11	Feature Engineering / Feature Selection	Features created or transformed, features removed, feature selection techniques, feature importance, and justification
12	Machine Learning Model Development	ML problem formulation, algorithms selected, rationale for algorithm selection, model implementation, training approach, and initial model results
13	Model Comparison	Comparison of at least three ML algorithms, performance comparison table, appropriate evaluation metrics, analysis of model differences, and selection of the best-performing model
14	Hyperparameter Tuning	Hyperparameters selected, tuning strategy, best hyperparameters, and comparison of performance before and after tuning
15	Final Model Evaluation	Final model performance, appropriate evaluation metrics, confusion matrix/plots where applicable, test-set results, and interpretation of results in the context of the problem
16	Model Strengths, Limitations and Improvements	Model strengths, limitations, dataset limitations, challenges, possible improvements, and future scope
17	Model Deployment	Deployment approach, Streamlit/Gradio/Flask implementation, application workflow/architecture, user inputs, prediction process, outputs, screenshots, and application usage
18	GitHub Repository	Repository link, project structure, source code, trained model, requirements.txt, README, installation instructions, and execution instructions The link to GitHub to be mentioned in the document (Source code is not required)
19	Conclusion	Summary of the problem, methodology, major findings, final model, performance, deployment outcome, and overall project outcome
20	References	Research papers, books, datasets, websites, documentation, libraries, and other sources used
21	Appendices	Additional graphs, tables, experimental results, screenshots, configuration details, or other supporting information
22	Final Approval Comments	Final approvals comments + recommendations from the guide